# Experiment 2

**Name: N. PARNITA** **Course Code: MLA0305**

**Regno: 192525322** **Slot: B**

**Title:** Implementation and Simulation of Markov Decision Process (MDP) for Grid World Navigation

**Software Required:**

- Google Colab
- Python
- NumPy
- Matplotlib

## AIM

To implement a Markov Decision Process (MDP) for Grid World Navigation and determine the optimal value of each state using Python.

## PROCEDURE

1. Open Google Colab.
2. Import the required libraries.
3. Define a Grid World environment.
4. Assign rewards to each state.
5. Apply Value Iteration to calculate state values.
6. Display the optimal values.
7. Compare the rewards and computed state values using a graph.
8. Observe the results.

## CODE:

```
initial values = np.zeros(9)
final_values = values.flatten()

states = [f"S{i+1}" for i in range(9)]
```

```
x = np.arange(len(states))
width = 0.35

plt.figure(figsize=(10,5))

plt.bar(x-width/2, initial_values, width, label="Initial Values")
plt.bar(x+width/2, final_values, width, label="Final Values")

plt.xticks(x, states)
plt.xlabel("Grid States")
plt.ylabel("State Value")
plt.title("Initial vs Final State Values using MDP")
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

## OUTPUT: